检测报告

编号 : GZMR240400842204_CN

日期 : 2024-05-11

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GZMR240400842204

报告验真请访问: check.sgsonline.com.cn

客户名称： 宁波哲能精密塑料有限公司

客户地址： 宁波市江北区姜湖路 406 号 5 幢 6 幢

样品名称 : PEEK

以上信息及样品由客户提供及确认，SGS 不承担证实客户提供信息的准确性、适当性和(或)完整性的责任。

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SGS 参考号 : NBIN2404000821PL04

收样日期 : 2024-04-01

测试开始时间 : 2024-04-01

测试结束时间 : 2024-05-11

测试结果 : 请见下页（除另有特别说明外，此报告结果仅对测试样品负责）

通标标准技术服务有限公司广州分公司
授权签名

霍洁梅

霍洁梅 授权签字人

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检测报告

结果总结：

序号	测试项目	测试方法	结果		结论
1	垂直燃烧	UL 94-2023 第 8 节	V-0		/
2	表面电阻	IEC 62631-3-2:2023	6.97E+13Ω		/
3	体积电阻	IEC 62631-3-1:2023	2.64E+14Ω		/
4	拉伸测试	ISO 527-2:2012	拉伸强度	96.3MPa	/
			断裂拉伸应变	4.4%	
5	弯曲测试	ISO 178:2019 方法 A	弯曲强度	170MPa	/
			弯曲模量	4010MPa	
6	洛氏硬度	ISO 2039-2:1987	113 L		/
7	比重	ISO 1183-1:2019 方法 A	1.301		/
8	吸水率	ISO 62:2008 方法 1	0.04%		/
9	平均线性热膨胀系数	ISO 11359-2:2021 方法 A	0℃~140℃	53×10^{-6} K^{-1}	/
			160℃ ~200℃	124×10^{-6} K^{-1}	
10	负荷变形温度	ISO 75-2:2013 方法 A	160℃		/
11	熔融温度(Tm)	ISO 11357-3:2018	详见结果		/
12	介电常数	IEC 62631-2-1:2018	3.204		/
13	击穿电压和电气强度	IEC 60243-1:2013	详见结果		/
14	简支梁冲击强度	ISO 179-1:2023	6.88kJ/m^2 C (完全破坏)		/

备注： 合格：达到要求

不合格：未达到要求

/：不下判定

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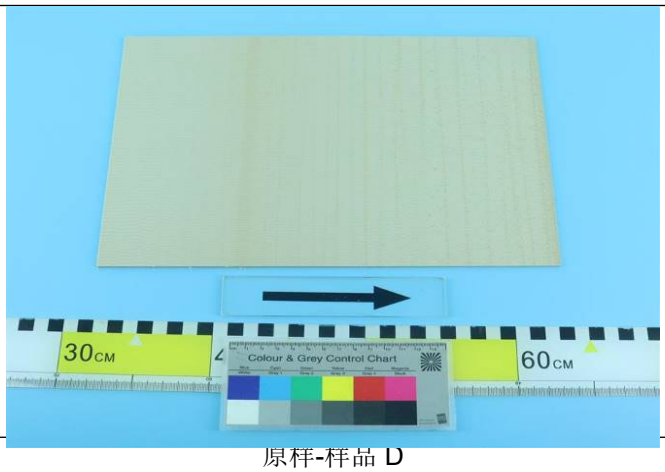
原样照片：



原样-样品 D

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1.测试项目：垂直燃烧

样品描述：片材

测试方法：UL 94-2023 第 8 节

测试条件：

试样：125mm×13.0mm×3.7mm

测试结果：

处理条件	常态：(23±2)℃，(50±5)%RH，48h					老化：(70±2)℃，168h→干燥器，(23±2)℃，4h				
试样	1	2	3	4	5	1	2	3	4	5
第一次施加火焰后余焰时间 (s)，t_1	0	0	0	0	0	0	0	0	0	0
第二次施加火焰后余焰时间 (s)，t_2	0	0	0	0	0	0	0	0	0	0
第二次施加火焰后余辉时间 (s)，t_3	0	0	0	0	0	0	0	0	0	0
任何状态调节，总余焰时间 (s)，t_1+t_2	0					0				
第二次施加火焰后余焰时间加余辉时间 (s)，t_2+t_3	0	0	0	0	0	0	0	0	0	0
试样余焰或余辉是否蔓延到夹具	否	否	否	否	否	否	否	否	否	否
燃烧颗粒或滴落物是否引燃脱脂棉	否	否	否	否	否	否	否	否	否	否

等级：V-0 (见表 1)

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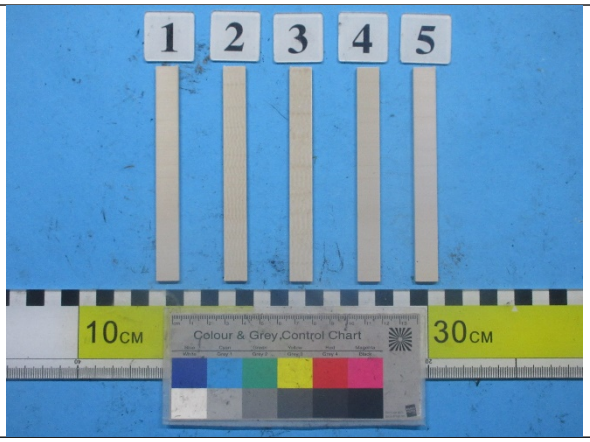
表 1

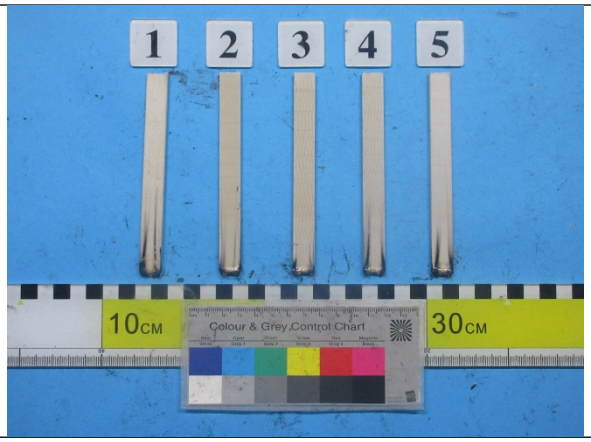
判定条件	V-0	V-1	V-2
单个试样余焰时间，t_1 和 t_2	≤10s	≤30s	≤30s
任何状态调节，总余焰时间(5 个试样的 t_1+t_2)	≤50s	≤250s	≤250s
第二次施加火焰后单个试样余焰时间加余辉时间 (t_2+t_3)	≤30s	≤60s	≤60s
任一试样余焰或余辉是否蔓延到夹具	否	否	否
燃烧颗粒或滴落物是否引燃脱脂棉	否	否	是

备注：测试试样从样品中裁取。

测试照片：



测试前-样品 D-常态



测试后-样品 D-常态

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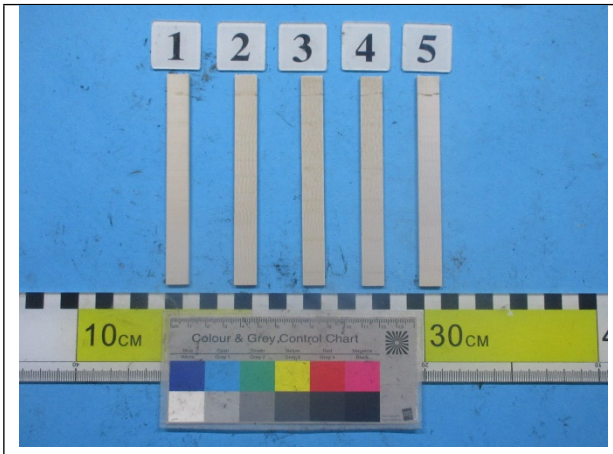	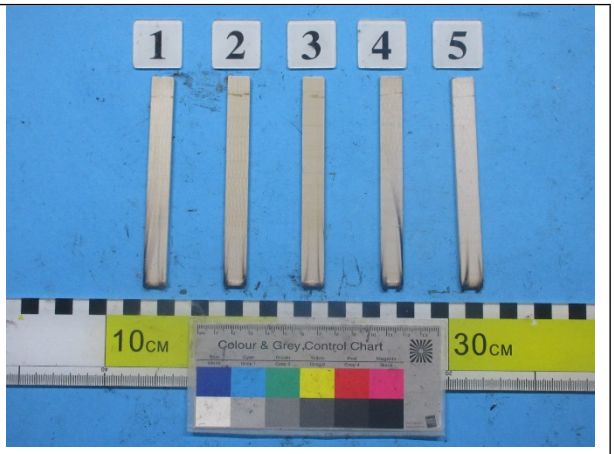
测试前-样品 D-热老化	测试后-样品 D-热老化

设备信息：

设备	型号	设备编号	校准日期	下次校准日期
水平垂直燃烧仪	HVUL2	GZMR-PL-E297	2023-04-20	2024-04-19
电热鼓风烘箱	UF750	GZMR-AG-E181	2023-08-24	2024-08-23

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2.测试项目：表面电阻

样品描述：片材

测试方法：IEC 62631-3-2:2023

测试条件：

预处理条件：(23±2) ℃，(50±5) %RH，96 h

测试电极： D_1=50 mm，D_2=70 mm

测试电压：100 Vdc

测试时间：1 min

实验室环境：(23±2) ℃，(50±5) %RH

测试结果：

试样编号	表面电阻 R_s (Ω)
1	4.56E+13
2	7.17E+13
3	2.10E+14
4	6.97E+13
5	4.93E+13
中值	6.97E+13

备注：

1) 测试试样从样品中裁取。

2) E+n 是指 10^n，例如 1.00E+03 等同 1.00×10^3。

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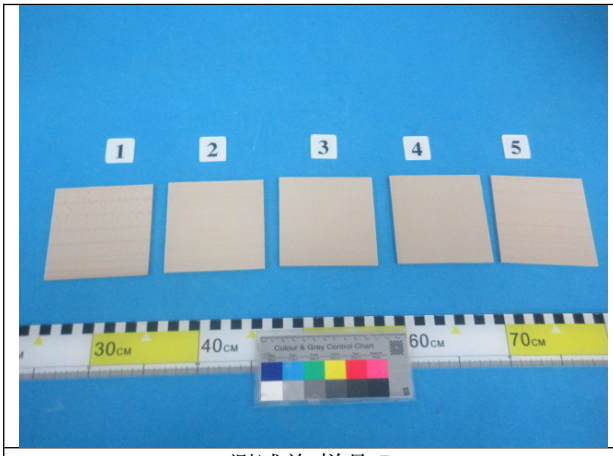	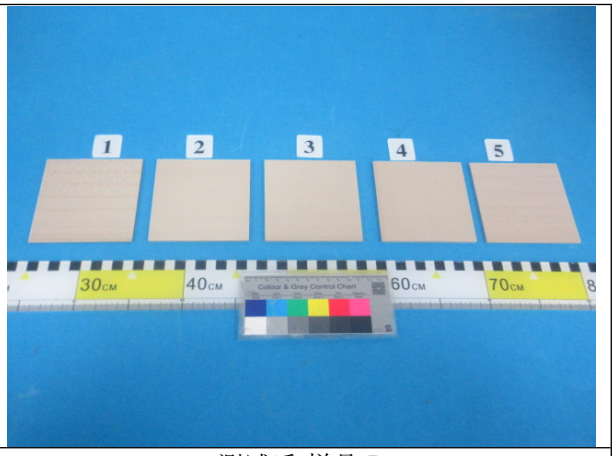
测试前-样品 D	测试后-样品 D

设备信息：

设备	型号	设备编号	校准日期	下次校准日期
高阻计	4339B	GZMR-PL-E150	2023-05-16	2024-05-15

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3.测试项目：体积电阻

样品描述：片材

测试方法：IEC 62631-3-1:2023

测试条件：

预处理条件：(23±2) ℃，(50±5) %RH，96 h

测试电极： D_1=50 mm，D_2=70 mm

测试电压：100 Vdc

测试时间：1 min

实验室环境：(23±2) ℃，(50±5) %RH

测试结果：

试样编号	厚度 (mm)	体积电阻 R_v (Ω)
1	3.69	2.07E+14
2	3.61	3.57E+14
3	3.74	2.64E+14
4	3.67	3.26E+14
5	3.64	8.79E+13
中值	/	2.64E+14

备注：

1) 测试试样从样品中裁取。

2) E+n 是指 10^n，例如 1.00E+03 等同 1.00×10^3。

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SGS-CSTC Standards Technical Services Co., Ltd. (Guangzhou Branch)

Guangzhou Branch Testing Laboratory

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中国・广东・广州高新技术产业开发区科学城科珠路198号 中国・广东・广州市黄埔区开创大道1501号3栋101房 t(86-20) 8215555 sgs.china@sgs.com

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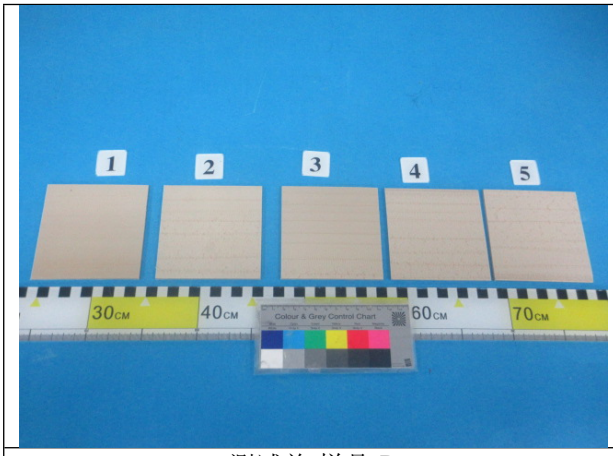
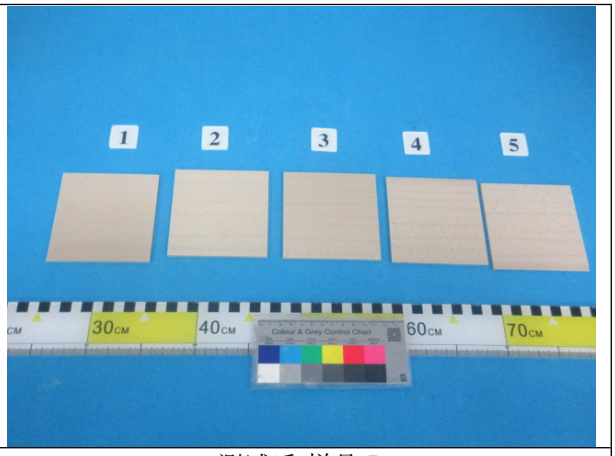
检测报告

编号 ： GZMR240400842204_CN

日期 ： 2024-05-11

测试照片：

	
测试前-样品 D	测试后-样品 D

设备信息：

设备	型号	设备编号	校准日期	下次校准日期
高阻计	4339B	GZMR-PL-E150	2023-05-16	2024-05-15
厚度计	S112MXB	GZMR-RL-E115-01	2023-10-31	2024-10-30

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4.测试项目：拉伸测试

样品描述：片材

测试方法：ISO 527-2:2012

测试条件：

试样类型：1B

试样厚度：3.67mm

测试速度：5mm/min

标距：50mm

初始夹具间距：115mm

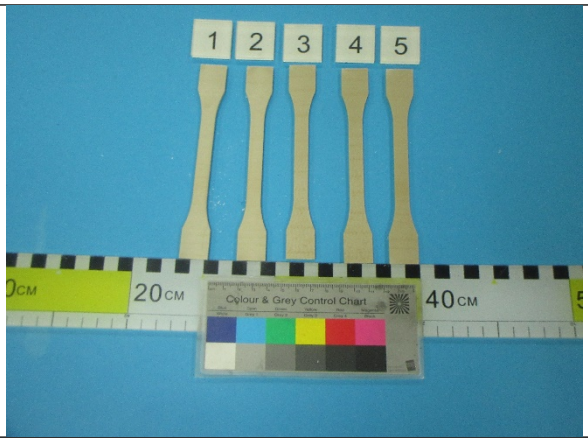
实验室环境条件：(23±2) ℃，(50 ± 5) % RH

测试结果：

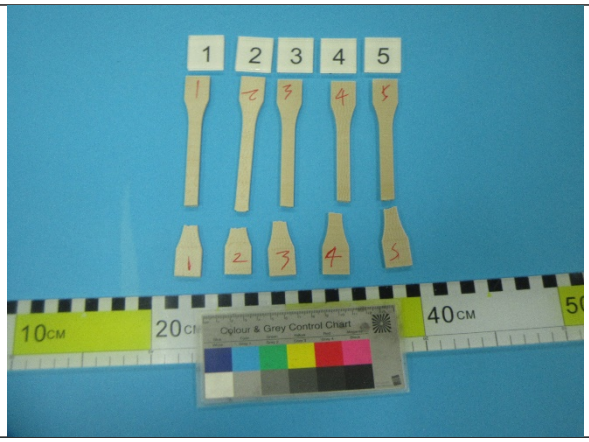
测试项目	测试结果
拉伸强度	96.3MPa
断裂拉伸应变	4.4%

备注： 测试试样从样品中裁取。

测试照片：



测试前-样品 D



测试后-样品 D

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设备信息：

设备	型号	设备编号	校准日期	下次校准日期
自动化电子万能材料试验机	Z020	GZMR-PL-E399	2023-05-15	2024-05-14

5.测试项目：弯曲测试

样品描述：片材

测试方法：ISO 178:2019 方法 A

测试条件：

试样：80mm×10.12mm×3.72mm

测试速度：2mm/min

跨距：59mm

实验室环境条件：(23±2)℃，(50±5)%RH

测试结果：

测试项目	测试结果
弯曲强度	170MPa
弯曲模量	4010MPa

备注： 测试试样从样品中裁取。

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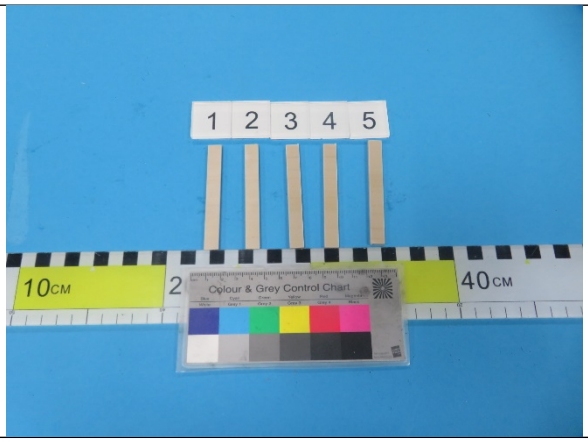
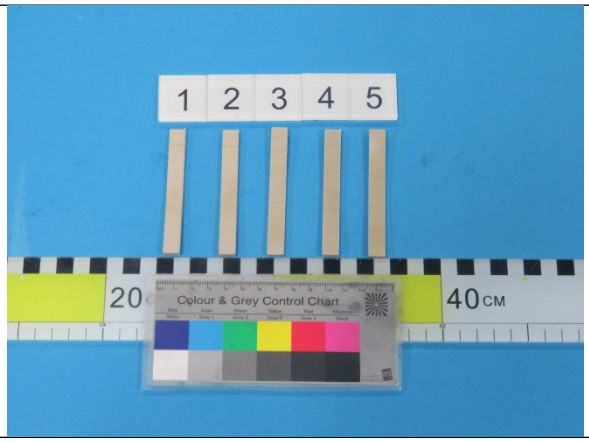
测试照片：

	
测试前-样品 D	测试后-样品 D

设备信息：

设备	型号	设备编号	校准日期	下次校准日期
电子万能材料试验机	Z1.0	GZMR-PL-E252	2023-11-01	2024-10-31

6.测试项目：洛氏硬度

样品描述：片材

测试方法：ISO 2039-2:1987

测试条件：

试样厚度：7.17mm (2 层叠加)

实验室环境条件：23 ± 2 ℃，50 ± 5 %RH

测试结果：

测试项目	测试结果
洛氏硬度	113 L

备注：测试试样从样品中裁取。

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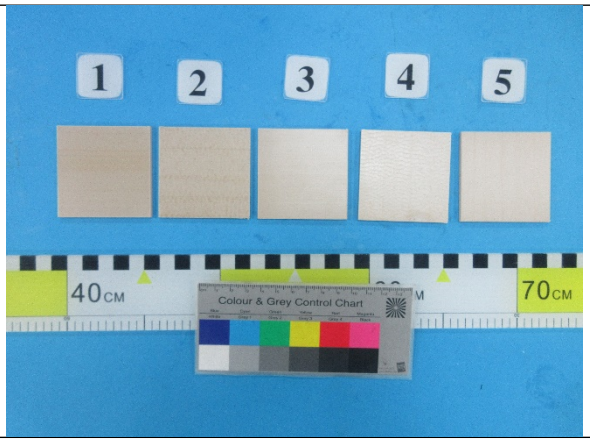
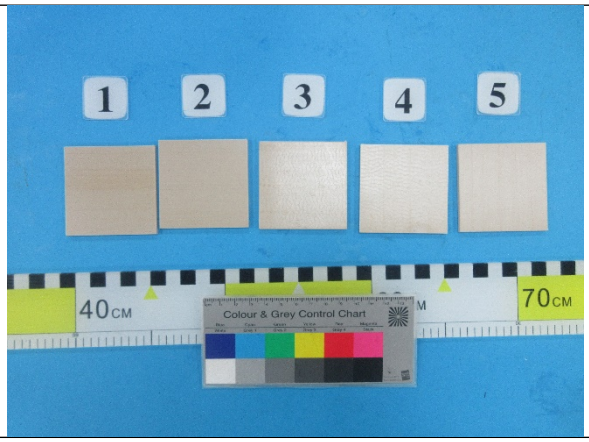
测试照片：

	
测试前-样品 D	测试后-样品 D

设备信息：

设备	型号	设备编号	校准日期	下次校准日期
洛氏硬度试验机	HR-521	GZMR-MT-E080	2024-01-04	2025-01-03

7.测试项目：比重

样品描述：片材

测试方法：ISO 1183-1:2019 方法 A

测试条件：

无水乙醇，(23±0.5)℃

实验室环境条件：(23±2)℃，(50±5)%RH

测试结果：

测试项目	测试结果
比重	1.301

备注：测试试样从样品中裁取。

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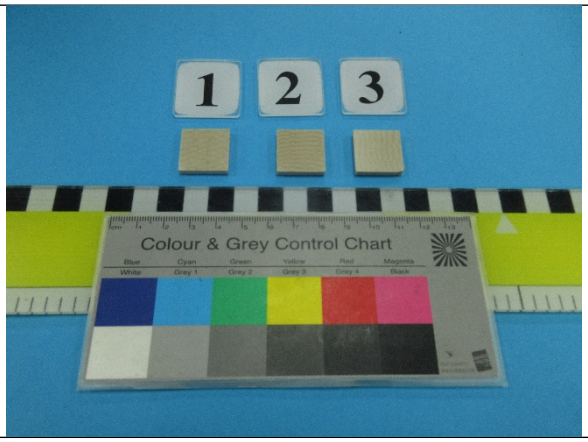
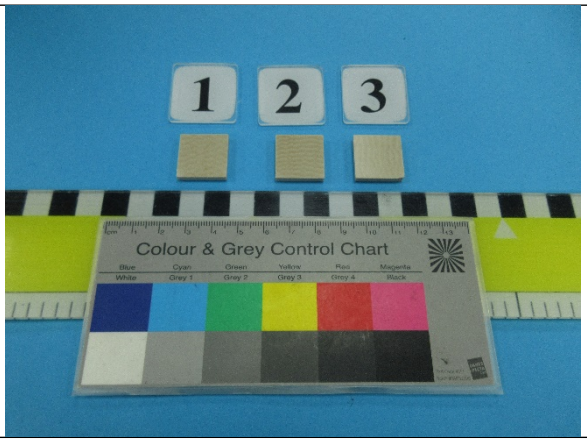
检测报告

编号 ： GZMR240400842204_CN

日期 ： 2024-05-11

测试照片：

	
测试前-样品 D	测试后-样品 D

设备信息：

设备	型号	设备编号	校准日期	下次校准日期
电子天平	XPE204	GZMR-PL-E004-06	2023-05-15	2024-05-14

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8.测试项目：吸水率

样品描述：片材

测试方法：ISO 62:2008 方法 1

测试条件：

试样：62mm×62mm×3.8mm

干燥条件：50±2℃，24h

浸泡条件：23±2℃，24h

实验室环境条件：(23±2)℃，(50±5)%RH

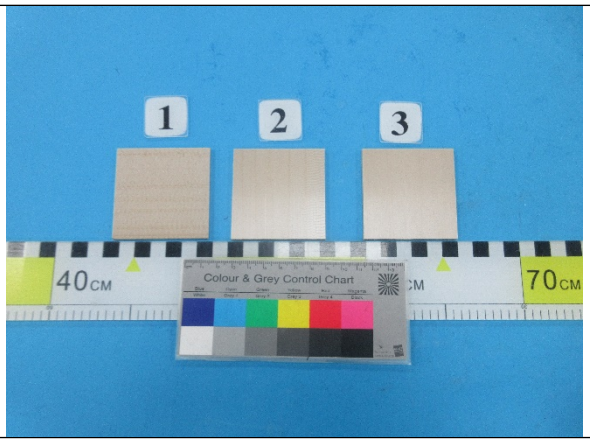
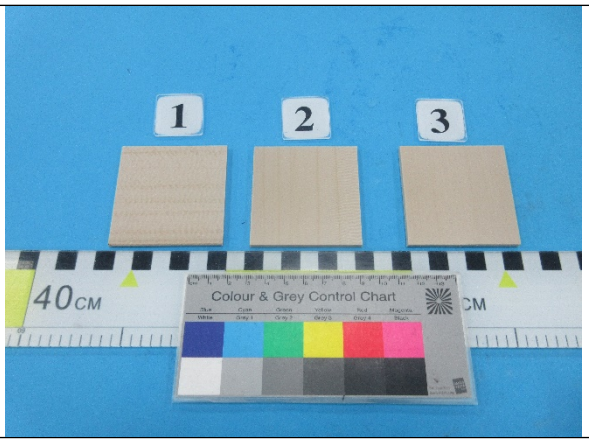
测试结果：

测试项目	测试结果
吸水率	0.04%

备注：

1. 吸水率，%=(浸泡后的质量-浸泡前干燥后的质量)/浸泡前干燥后的质量×100。
2. 测试试样从样品中裁取。

测试照片：

	
测试前-样品 D	测试后-样品 D

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设备信息：

设备	型号	设备编号	校准日期	下次校准日期
电子天平	XPE404S	GZMR-PL-E004-07	2023-08-11	2024-08-10
自然换气烘箱	UN110	GZMR-PL-E265	2023-05-12	2024-05-11

9.测试项目：平均线性热膨胀系数

样品描述：片材

测试方法：ISO 11359-2:2021 方法 A

测试条件：

试样：9.97mm×5.43mm×3.42mm

升温速率：5 ℃/min

负载：50mN

气体流动速率(N_2)：50 ml/min

测试温度：0 ℃~200 ℃

测量模式：压缩

测试方向：长度

试样数：1

实验室环境条件：(23 ± 2) ℃，(50 ± 5) % RH

测试结果：

测试项目	测试结果	
平均线性热膨胀系数	0℃~140℃	53×10^{-6} K^{-1}
	160℃~200℃	124×10^{-6} K^{-1}

备注：

1.测试试样从样品中裁取。

2. 测试的结果取自退火处理后。

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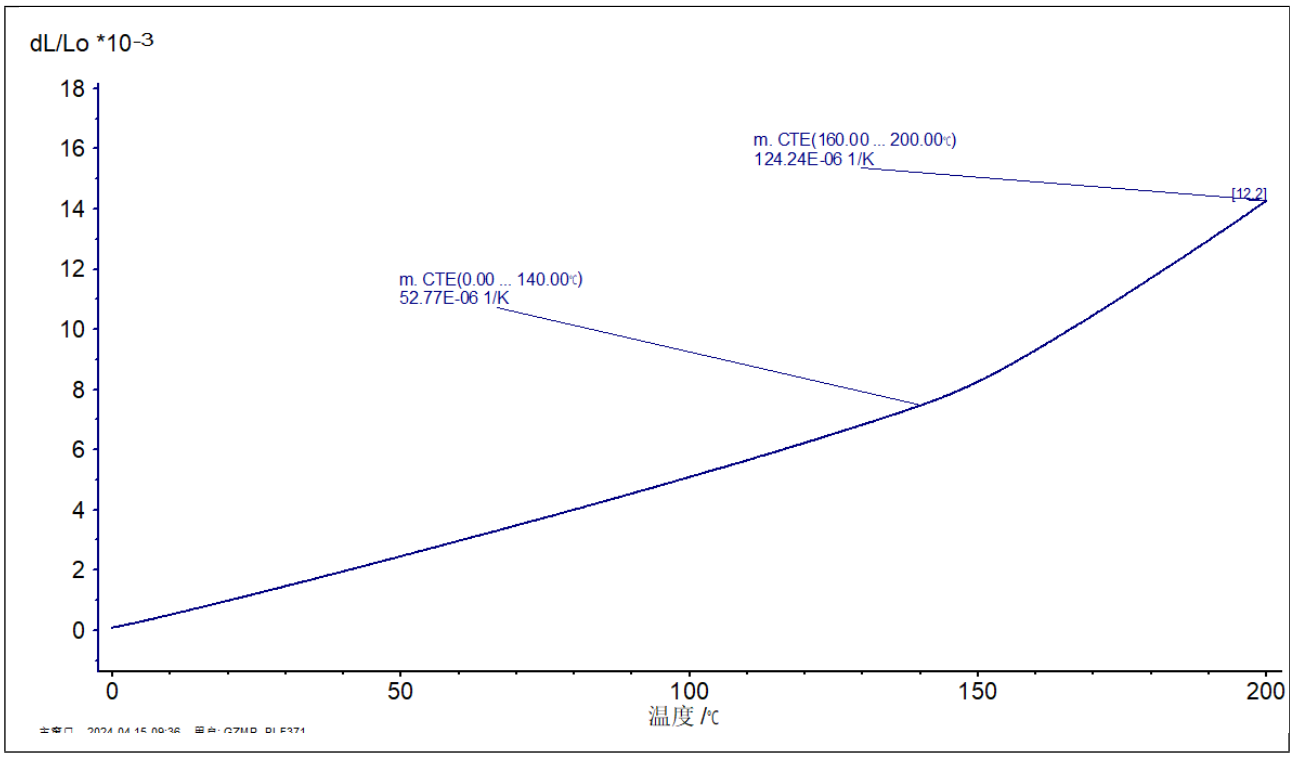
曲线图-长度方向：



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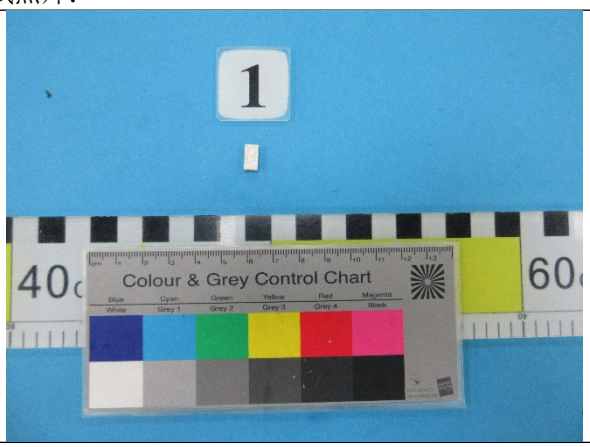
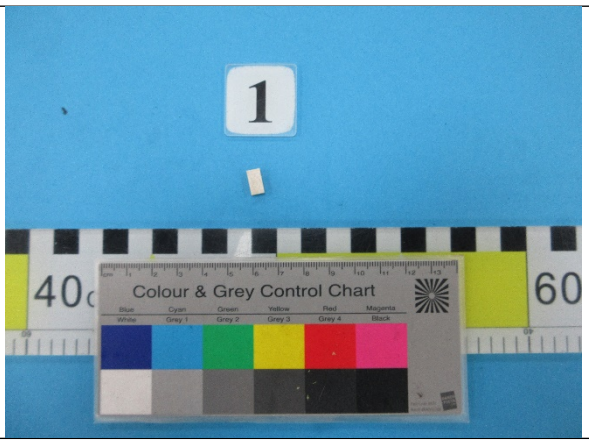
检测报告

测试照片：

	
测试前-样品 D	测试后-样品 D

设备信息：

设备	型号	设备编号	校准日期	下次校准日期
热机械分析仪	TMA402F3	GZMR-PL-E371	2023-06-06	2024-06-05

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10. 测试项目：负荷变形温度

样品描述：片材

测试方法：ISO 75-2:2013 方法 A

测试条件：

试样：80mm×10.22mm×3.74mm

传热介质：硅油

T_f1.8

升温速率：120℃/h

跨距：64mm

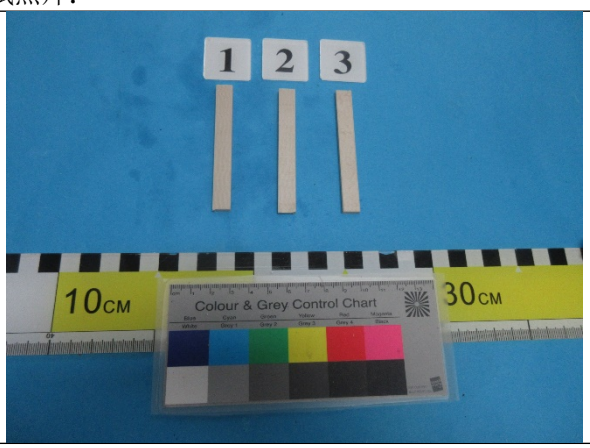
实验室环境条件：23 ± 2 ℃，50 ± 5 % RH

测试结果：

测试项目	测试结果
负荷变形温度	160℃

备注：测试试样从样品中裁取。

测试照片：

	
测试前-样品 D	测试后-样品 D

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设备信息：

设备	型号	设备编号	校准日期	下次校准日期
热变形维卡测定仪	IC 6	GZMR-PL-E348	2023-06-21	2024-06-20

11.测试项目：熔融温度(T_m)

样品描述：片材

测试方法： ISO 11357-3:2018，分析仪器为 DSC。

测试条件：

样品容器类型：Al；

吹扫气体流速(氮气：99.999%)：50ml/min；

升温范围：40~380℃；

升温速率：10℃/min。

测试结果：

测试项目	第一次升温扫描 (DSC 曲线见图 1)	第二次升温扫描 (DSC 曲线见图 2)
外推熔融起始温度 T_{eim}	320.6℃	327.4℃
熔融峰温 T_{pm}	340.8℃	341.1℃
外推熔融终止温度 T_{efm}	348.4℃	347.6℃

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检测报告

编号 ： GZMR240400842204_CN

日期 ： 2024-05-11

图 1

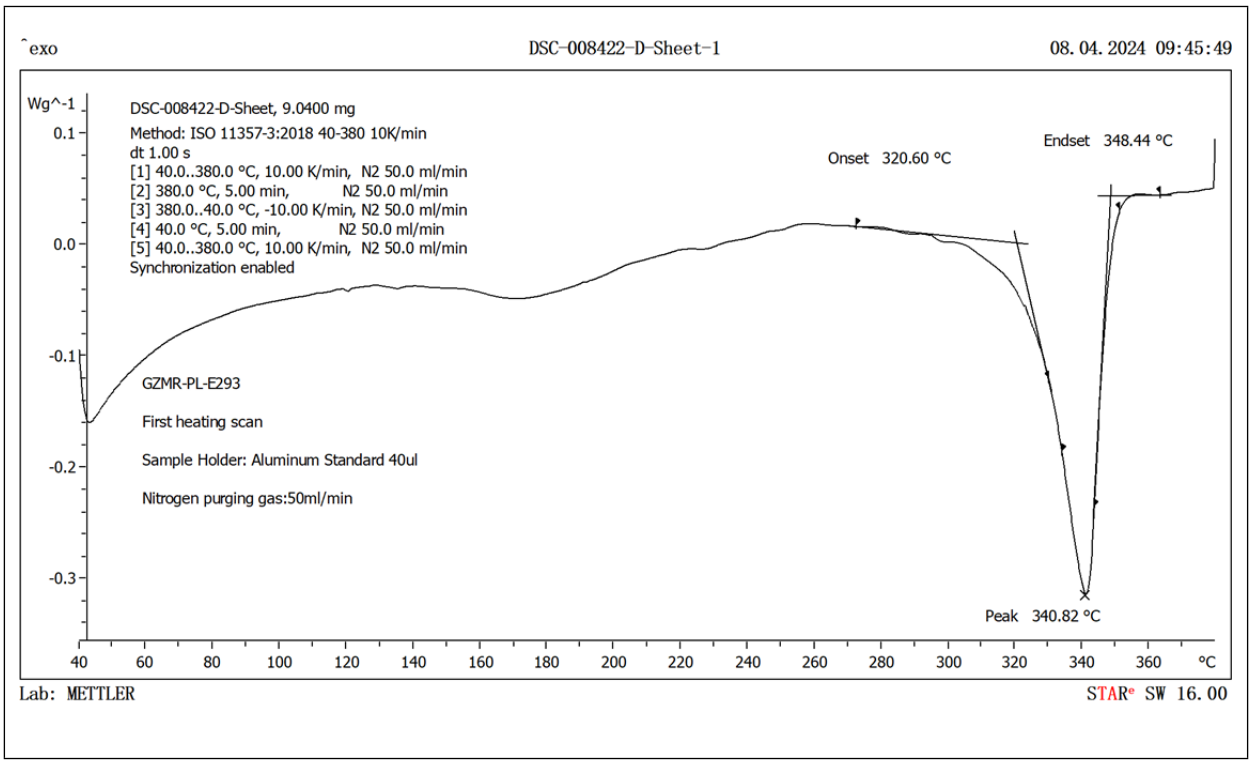

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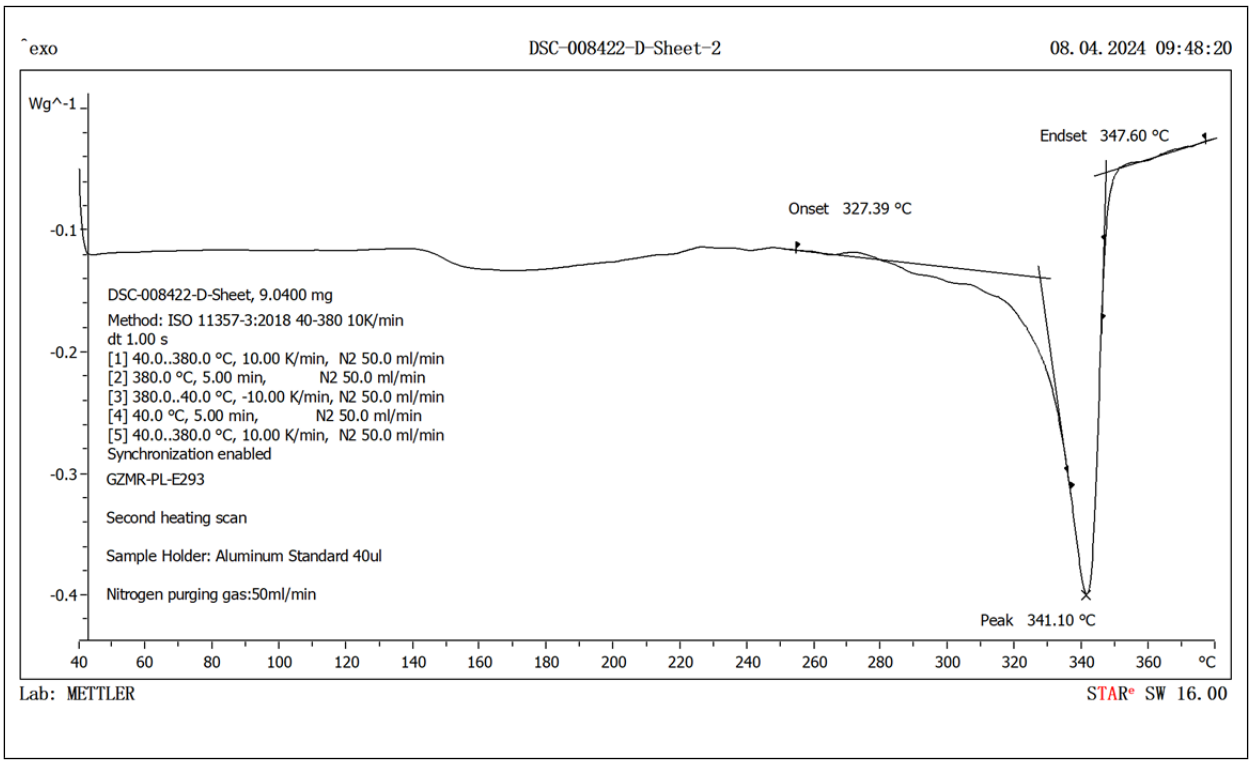
图 2



设备信息：

设备	型号	设备编号	校准日期	下次校准日期
差示扫描量热仪	DSC 3	GZMR-PL-E293	2024-03-05	2026-03-04

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12.测试项目：介电常数

样品描述：片材

测试方法：IEC 62631-2-1:2018

测试条件：

预处理条件：(23±2) ℃，(50±5) %RH，96 h

测试电极：平行电极 φ38 mm

测试频率：1 M Hz

实验室环境：22.3 ℃，51.6 %RH

测试结果：

试样编号	厚度 (mm)	电容 C_p (pF)	介电常数 ε_r	损耗因数 D
1	3.50	9.217	3.214	0.00315
2	3.71	8.684	3.210	0.00304
3	3.84	8.348	3.194	0.00298
4	3.86	8.312	3.197	0.00294
5	3.40	9.464	3.206	0.00327
平均值	/	/	3.204	0.00308

备注：

1.测试试样从样品中裁取。

2.此测试项目是由通标标准技术服务(上海)有限公司执行的。

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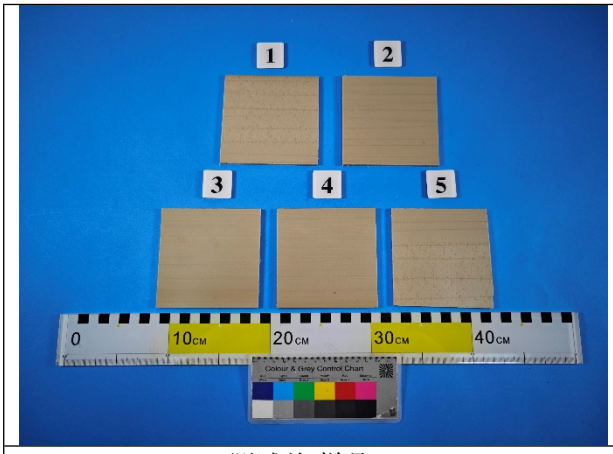
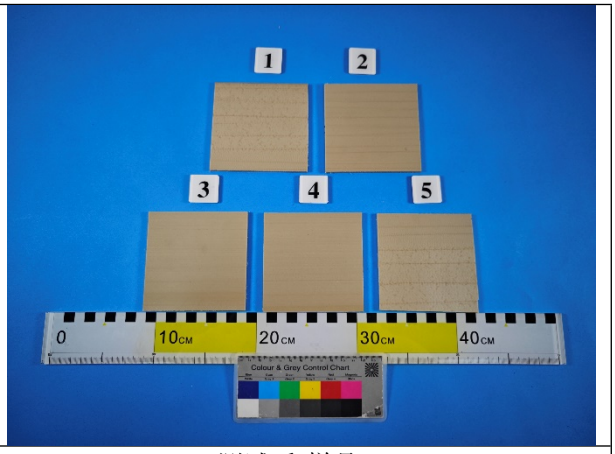
检测报告

编号 ： GZMR240400842204_CN

日期 ： 2024-05-11

测试照片：

	
测试前-样品 D	测试后-样品 D

设备信息：

设备	型号	设备编号	校准日期	下次校准日期
LCR 测试仪	E4980AL	P-576	2023-10-16	2024-10-15
电介质测试夹具	16451B	P-576-1	2023-10-17	2024-10-16
数显卡尺	0-150mm-0.01mm	P-698	2024-03-01	2025-02-28

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13 测试项目：击穿电压和电气强度

样品描述：片材

测试方法：IEC 60243-1:2013

测试条件：

预处理条件：(23±2) ℃，(50±5) %RH，24 h

周围媒质：(23±2) ℃变压器油

测试电极：不等直径电极(上电极 ϕ25 mm、下电极 ϕ75 mm)

电压类型：交流 50 Hz

升压方式：快速升压，2000 V/s

测试结果：

试样编号	厚度 (mm)	击穿电压(kV)	电气强度(kV/mm)
1	3.68	37.99	10.32
2	3.67	37.43	10.20
3	3.69	37.12	10.06
4	3.68	37.62	10.22
5	3.67	37.69	10.27
中值	--	37.62	10.22

备注：测试试样从样品中裁取。

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Guangzhou Branch Testing Laboratory

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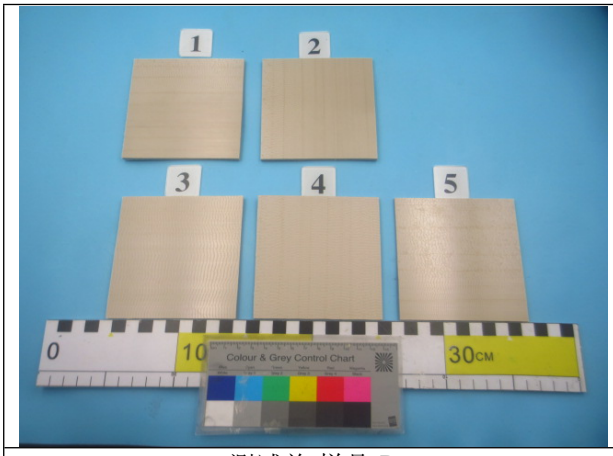
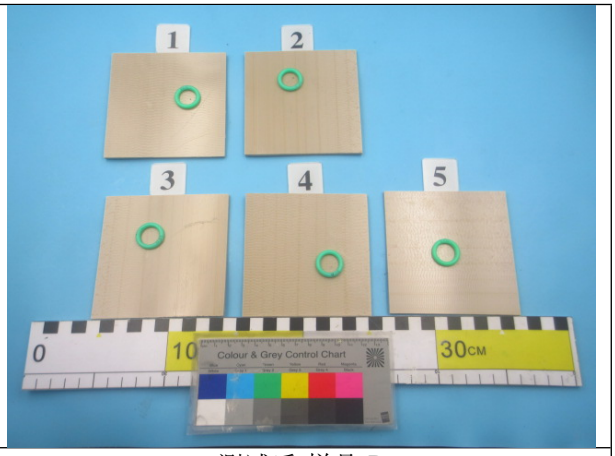
测试照片：

	
测试前-样品 D	测试后-样品 D

设备信息：

设备	型号	设备编号	校准日期	下次校准日期
微机控制电压击穿试验仪	DJC-100KV	GZMR-PL-E154-01	2023-06-07	2024-06-06
厚度计	S112MXB	GZMR-RL-E115-01	2023-10-31	2024-10-30

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SGS-CSTC Standards Technical Services Co., Ltd. (Guangzhou Branch) 检验检测专用章 Inspection & Testing Services

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14.测试项目：简支梁冲击强度

样品描述：片材

测试方法：ISO 179-1:2023

测试条件：

试样：ISO 179-1/1eA (缺口制备：机械加工)

试样厚度：3.72mm

摆锤能量：1J

冲击速度：2.9m/s

跨距：62mm

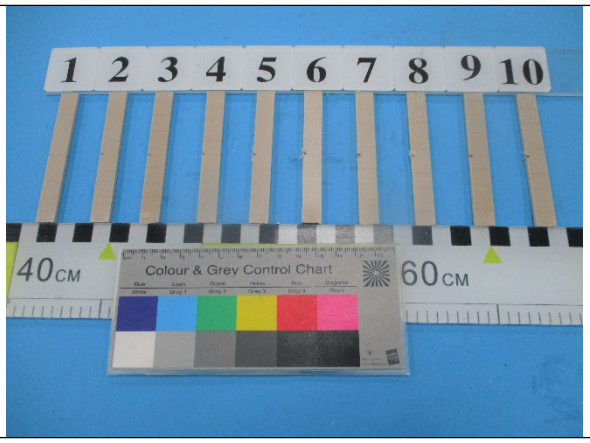
实验室环境条件：(23±2) ℃，(50±5) %RH

测试结果：

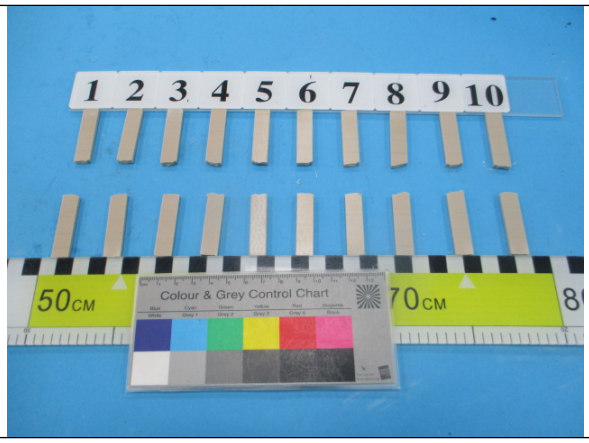
测试项目	测试结果
简支梁缺口冲击强度	6.88kJ/m^2 C (完全破坏)

备注： 测试试样从样品中裁取。

测试照片：



测试前-样品 D



测试后-样品 D

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检测报告

设备信息：

设备	型号	设备编号	校准日期	下次校准日期
冲击试验仪	HIT50P	GZMR-PL-E283	2023-06-26	2024-06-25

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